

65 kg.

2 Ball A is projected horizontally at 2.0 m s^{-1} from the top of a vertical cliff while Ball B is released from rest 1 s later from the same point. It took Ball B 3.5 s to reach the base of the cliff. How far from the base of the cliff will Ball A hit the ground?

A 7.0 m

B 9.0 m

C 53 m

D 67 m

Ans: A

Both A and B will take the same amount of time and thus 3.5 s. Hence, horizontal distance = $3.5 \times 2 = 7.0 \text{ m}$